

Lily Harper Holmes

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Education

University of Colorado Boulder

M.S. in Data Science

Boulder, CO

Expected May 2027

Relevant Coursework: Statistical Methods I & II, Data Structures & Algorithms, Data Mining

GPA: 3.94/4.00

Coursework in Progress: Machine Learning, Natural Language Processing, Modern Big Data Tools

University of Nebraska Lincoln

Lincoln, NE

B.A. in Economics and History, minor in Art History

May 2024

Relevant Coursework: Introductory Econometrics, Economic Data Visualization & Analysis,

GPA: 3.77/4.00

Applied Game Theory, Labor Economics, History Capstone

with Distinction

Skills

Programming & Data: Python, R, SQL, pandas, scikit-learn, NumPy, matplotlib, seaborn, tidyverse, ggplot2

Analysis & Modeling: Regression, classification, causal inference, exploratory data analysis, model evaluation

Tools: Excel, Tableau, Power BI, Git, GitHub, Claude Code, Codex

Communication: Technical writing, data visualization, statistical interpretation

Projects

Denver Motor Vehicle Injury Risk Classifier *Python, pandas, scikit-learn*

July 2026

- Built a reproducible Python pipeline with 25 engineered features on Denver open crash data (270K+ reports, 2013–2026) to predict serious-injury or fatal crashes, a 2.2% positive class.
- Compared logistic regression thresholds, decision tree, naive Bayes, and baseline models with consideration toward asymmetric error costs, prioritizing the reduction of false-negatives.
- Held out 2025–26 crashes as an untouched test set and chose the model and threshold on 2024 data. On the test set the model caught 69% of severe crashes (317 of 458) at ROC-AUC 0.85 and PR-AUC 0.23, about 10× the 0.02 baseline.

Undergraduate Thesis on Nudges and Student Decision-Making *R, tidyverse, ggplot2*

May 2024

- Secured IRB approval and partnered with College of Business deans to evaluate a student-retention program using sensitive, de-identified enrollment records. Presented results, assumptions and limitations in a thesis and oral defense.
- Cleaned de-identified records for 3,700+ student observations across the 2021 and 2022 campaigns and found no consistent program effect across years.
- Built treatment groups (reached by phone, left a message, not contacted), call-timing and caller-role variables, and estimated linear probability models in R with controls for academic level, GPA and year.

Experience

Polco

Boulder, CO

Data Specialist (Contract)

Oct 2025 – Dec 2025

- Researched, categorized, and validated data in Excel for a transportation-planning project supporting Boulder's Transportation Master Plan.
- Supported stakeholder-facing analysis requiring careful documentation and repeatable data-validation workflows.

Methodology and Evaluation Research Core Facility

Lincoln, NE

Undergraduate Research Assistant

June 2023 – Sept 2023

- Analyzed survey data and created visualizations for evaluation reports used by internal and external stakeholders.
- Generated descriptive statistics and exploratory findings using IBM SPSS and Excel.
- Translated raw survey responses into concise findings for research and reporting teams.

College of Business Teaching and Learning Center

Lincoln, NE

Teaching Assistant / Learning Assistant

Jan 2023 – Dec 2023

- Explained quantitative, statistical, and economic concepts to students with varied levels of technical preparation.
- Served as a tutor for multiple introductory in-person and online economics courses (introductory micro/macroeconomics) with up to 300 students/section, leading review sessions, one-on-one, and group tutoring.
- Served as the Teaching Assistant for an upper-level economics course (Labor Economics), graded assignments and exams and supported exam administration.

Certifications

[AWS AI Practitioner](#)